Module 2

Hello and welcome to TEK 100! I'm Katie Thompson

And I'm Tama Thé. Welcome to Module 2: The AI Co-Pilot: Augmenting Your Ideas

Introduce core theme of the module: In Module 1, you learned to *spot* AI in the world around you. Now, in Module 2, we're going to start learning the best strategies and techniques to use AI as a collaborative partner. This is where we get into the meat of how to interact with large language models and generative AI. These are the tools that we use every single day in our own practice and research. In fact, a lot of this course was designed in conversation with generative AI to make sure that we are providing the most up-to-date and evidence-based recommendations on artificial intelligence.

Katie, you built this module for us, tell us about it.

- jump right in to using AI effectively!
- We'll focus on working **with** generative AI tools to get better results, better ideas, faster than if we work in isolation.
- Think of this like how we have conversations with other humans
  - Been thrown off by a question without context?
  - Been misinformed by thinking you are talking to someone in one role, only to find out their expertise, and therefore perspective/response, is not helpful?
  - This module focuses on HOW to converse with AI, in a framework called prompt engineering. That's a fancy way of saying AI communication skills, just like current job postings often list "oral communication skills" as a requirement.

[Begin opening lecture]

This week, you'll learn one of the most important skills in working with AI:

how to write **effective prompts.** A prompt is simply what you type or say to the AI — but not all prompts are equal. Some are vague and get you generic answers.

AI is a collaborator, not a mind reader.

Large language models don't "know" what you want; they generate responses based on probabilities from their training data. A vague or poorly phrased prompt produces vague or unhelpful answers. A clear, structured prompt guides the AI to deliver more relevant, accurate, and usable output.

Iteration improves results.

The first answer from an AI is rarely the best. Prompt engineering teaches you how to **refine and iterate**, adding context, format, and constraints so the AI can align better with your goals. This iterative process is core to the TEK-100 course design, where students practice improving outputs step by step.

It helps humans stay "in the loop."

emphasize that humans must provide structure and critical oversight, because AI doesn't inherently reason or validate facts. A well-designed prompt adds the missing structure—like asking for step-by-step reasoning, or examples in a specific style—so that the AI's output is closer to what a human can trust and evaluate.

It's foundational for all higher skills.

In TEK-100, prompt engineering (Module 2: *The AI Co-Pilot*) is the first practical skill students learn because it underpins everything else—critical inquiry, fact-checking, multi-modal creativity, and project design. Without solid prompting, every later use of AI (research synthesis, creativity, workflow design) will be weaker.

So let's get into the module with some videos about how to engage with AI.

Main lecture - Basics of Prompt Engineering

When it became clear that crafting a strong prompt for generative AI was a skill of itself, there was a lot of buzz in the media about prompt engineering.

Prompt engineering is essentially **programming a machine with words**.

The quality of every response you get from an AI, from a simple summary to a complex analysis, hinges almost entirely on the quality of your instructions. This is the essence of prompt engineering: the skill of crafting clear, specific, and strategic inputs to get the most useful and accurate outputs.

Think of it like ordering at a coffee shop. If you walk in and just say, "I'll have a coffee," you'll get something generic. But if you give a detailed order—"a large iced latte with oat milk and one pump of vanilla"—you get exactly what you want. A well-crafted prompt works the same way, providing the AI with the specific details it needs to move beyond a generic response and deliver a tailored, high-quality result.

Mastering this skill is what separates a **passive user from an active collaborator**. It is the foundational technique that allows you to **guide the AI, turning it from a simple tool into a genuine "co-pilot" for your work**. By learning how to provide clear context, define a specific task, and ask for a precise format, you move into the driver's seat, ensuring that the AI is working to augment your goals and enhance your ideas, not just giving you the first answer it finds.

The most well-described and commonly used framework for crafting good prompts uses four pillars: role, task, context, and format.

1. **Role** – Who should the AI pretend to be?
    - Example: *"You are a career coach helping a college freshman explore majors."*
2. **Task –** What do you want it to do?
    - Example: *"List 3 possible majors based on their interests in biology and a love of the outdoors."*

3. **Context** – What background details help it do better?
    - Example: *"The student has a strong math background but struggles with reading for more than 5-10 minutes."*
4. **Format** – How should the answer be delivered?
    - Example: *"Write the answer as a table comparing pros and cons of each major."*

Now, we're going to get into each of them in much more detail.

1. Role → Biasing the Model's Output Distribution

- **What's happening under the hood:**

When you assign a role ("You are a career coach"), you're seeding the model with *contextual priors*. LLMs are probability engines — they generate the "most likely next word" given your input. If the system sees "You are a doctor," it biases toward medical vocabulary and authoritative tone.

- **Why it matters:**

Without role-setting, the model defaults to generic patterns. With role-setting, you constrain the probability space so it leans toward the *register, tone, and knowledge domain* you want.

2. Task → Reducing Ambiguity in the Loss Landscape

- **What's happening under the hood:**

LLMs were trained on a huge mix of text — Q&A, Wikipedia, Reddit threads, novels, recipes. If you just say "Tell me about dogs," the model doesn't know whether you want *a Wikipedia-style definition, a casual story, or a training manual*. By specifying the task, you collapse that ambiguity.

- **Why it matters:**

Less ambiguity = fewer irrelevant completions. The model focuses its sampling on a narrower set of outputs, making the response more accurate and less rambling.

3. Context → Conditioning on Relevant Tokens

- **What's happening under the hood:**

LLMs don't "know" your situation unless you encode it in text. Context tokens (audience, goals, constraints) are literally part of the sequence the model attends to in its transformer layers. They shift which parts of its massive training data it draws from.

Potty Training

- **Why it matters:**

The more context you provide, the more the model can pattern-match to training examples that resemble *your scenario*. Context effectively narrows the model's "search space" and increases coherence with your needs.

4. Format → Controlling Output Structure

- **What's happening under the hood:**

Asking for a table, bullet list, or script sets *expectations*. The model has seen countless examples of these formats in training data. Once you specify a format, it locks into those learned structures (e.g., "- " triggers bullet lists, "|" triggers tables).

- **Why it matters:**

This reduces cognitive load for the human — you don't have to restructure a long essay into the format you need. It also reduces hallucination risk because structural constraints act like "guardrails" on generation.

- Learn through audio (ppl need audio summary), written (narrative), visual

To practice we have built a custom GPT. It is an interactive exercise that will guide you step-by-step through the **Four Pillars of Prompting (Role, Task, Context, and Format)** to see firsthand how clear, layered instructions can transform a simple idea into a powerful and precise result.

After you get done with the assignment, we will meet back up with you on the

Discussion Board.

Beyond the Basics of The Four Pillars of Prompt Engineering

So we've now covered the foundational **Four Pillars of Prompting**—Role, Task, Context, and Format. Think of these as the "basic grammar" of talking to AI: they make sure your request is clear and structured.

But just like learning to write a good essay doesn't stop with grammar, effective prompting doesn't stop with the Four Pillars.

Working with AI is more like an **ongoing conversation** than a one-time command. The best results often come when you refine, experiment, and even "teach" the AI how to work with you. That's where these next strategies come in. Let's get into the more sophisticated ways to engineer your prompts.

Iteration & Refinement

ChatGPT is the best first draft generator. Students should learn to *revise their prompts*, ask follow-up questions, and experiment with variations.

This builds the skill of *conversational prompting*—treating the AI like a dialogue partner, not a vending machine.

Strategies:

Ask: *"Can you expand on point 3?"* or *"Give me a different angle."*

Rerun the same question with small tweaks (e.g., change audience, format, tone).

Keep a short log of at least 3 iterations to see improvement.

Specificity & Constraints

Vague prompts = vague answers. Adding constraints (word count, style, audience, perspective) helps the model generate sharper, more useful responses.

Strategies:

Use numbers: “Give me 3 examples,” “Summarize in 150 words.”

Define audience: “Explain for a high school science fair.”

Set style constraints: “Use bullet points,” “Make it persuasive.”

Stepwise Reasoning (Chain-of-Thought)

AI often struggles when asked to “jump to the conclusion.” Prompting it to *show steps* (“Explain your reasoning step by step”) increases accuracy.

NBME Grading analogy

Strategies:

Add: *“Explain step by step.”*
Break big tasks into sub-questions: instead of “Write my essay,” start with “Brainstorm 5 thesis statements,” then “Outline supporting points.”

Use correction: If it jumps to the end, prompt: *“Slow down, what’s step 1?”*

Examples & Demonstrations (Few-Shot Prompting)

Providing a model example of the desired output helps the AI imitate the structure or tone.

Strategies:

Share a mini example: “Here’s a good summary. Follow this structure.”

Give both good and bad samples, then ask it to imitate or improve.

When formatting: paste a short mock-up and say, *“Fill in this template with new info.”*

Error Awareness & Fact-Checking

Students should expect mistakes (bias, hallucinations, noise). Teach them to build *validation* into prompts (e.g., “List your sources,” “Check your answer against three perspectives”).

Strategies:

Prompt: *"List your sources"* or *"Double-check that against three perspectives."*

Cross-check with Google Scholar, Wikipedia, or trusted databases.

Use AI itself: *"What could be wrong with your last answer?"*

Iterative Role Shifts

Beyond "role" as one of the Four Pillars, students should learn how changing the AI's role mid-conversation can unlock new perspectives

Strategies:

Meta-Prompting (Prompting About Prompts)

Nobody can talk to AI like AI can talk to itself.

Ask the AI to *help improve your prompt*. For example: "Suggest three ways to rephrase this prompt to get clearer results."

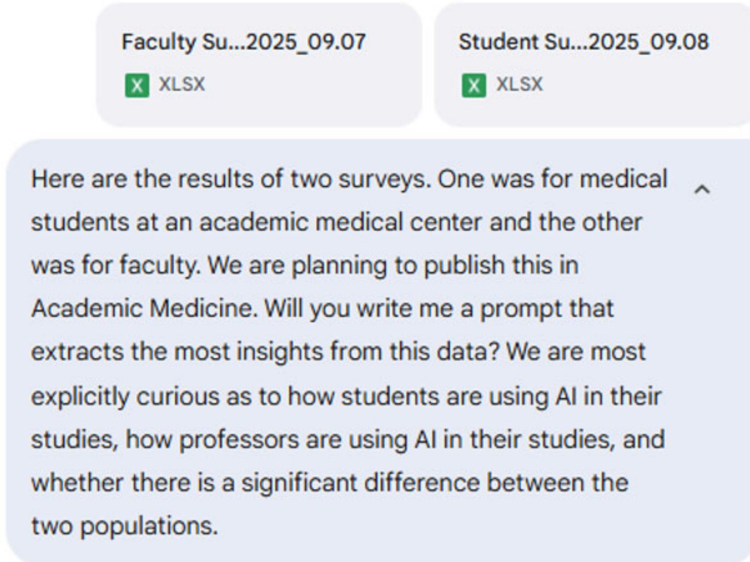
Example of write me a prompt

Strategies:

Ask: *"How could I phrase this prompt better?"*

Try: *"Suggest 3 improved versions of my question, one casual, one professional, one technical."*

Reflect: Compare the results of your original vs. AI-optimized prompt.



https://gemini.google.com/u/2/app/f79eea3ac164f5c4 - Prompt

https://gemini.google.com/u/2/app/573b5fc8945f7e8c - Report

Awareness of Input/Output Modalities

Different models respond differently depending on input (text vs. image) and output expectations (summary, chart, narrative). Being clear about the data type helps match the right tool.

Strategies:

Identify: “My input is text, my output should be an image” → use image generator.

Ask AI: *“What’s the best tool for this input/output?”*

Practice rephrasing: Convert one task into multiple forms (e.g., text summary, infographic outline, narrated script).

Ethical Guardrails

Prompts themselves can encode bias or unsafe requests. Teach students to recognize when they’re asking something problematic, and to reframe responsibly.

Strategies:

Do a “gut check”: *Would I be comfortable turning this in or publishing it* *under my name?*

Reframe prompts: Instead of *“Write my essay,”* → *“Help me outline key arguments.”*

Ask AI directly: *“What ethical issues might be raised by this request?”*

Practice rewriting prompts that could cross lines (privacy, fairness, plagiarism).

- define success and being thoughtful about the process - don't let AI do this one